

Ma'at as Constitutional Infrastructure: Why Advanced AI Systems Require a Ma'at-Governed Foundation

Abstract

This dissertation advances a normative and architectural thesis: all serious AI infrastructure should be governed by Ma'at, or by a framework functionally equivalent to Ma'at's demands for truth, balance, order, justice, reciprocity, and accountability. The claim is not that Ma'at is the only conceivable ethical vocabulary for AI. The claim is stronger and more practical: without a Ma'at-like constitutional layer, advanced AI systems predictably drift toward opacity, overreach, fragility, and moral incoherence. Current AI governance frameworks already recognize the need for transparency, human oversight, continuous evaluation, provenance, and risk management. NIST's AI Risk Management Framework and its Generative AI Profile treat AI risk as a lifecycle problem requiring governance, mapping, measurement, and management; UNESCO insists ultimate responsibility remains with humans and that monitoring be continuous and proportionate to risk; OECD emphasizes human agency, oversight, transparency, and explainability; OWASP now catalogs prompt injection, insecure output handling, training data poisoning, supply-chain risk, excessive agency, and overreliance as routine security concerns in LLM systems. In parallel, major frontier developers are explicitly treating AI as a critical-infrastructure security issue.

The dissertation argues that these modern requirements are best understood not as isolated compliance checkboxes but as fragments of a deeper governance logic that Ma'at already names with unusual precision. In both mainstream Egyptological summary and African-centered ethical scholarship, Ma'at denotes truth, order, balance, and justice; Maulana Karenga explicitly treats it as a moral ideal capable of contributing to modern ethical discourse. Read this way, Ma'at is not mere symbolism. It is a design grammar for systems that must remain coherent under pressure. The central conclusion, then, is architectural: AI should not merely be aligned at the model layer; it must be constituted at the infrastructure layer. That constitution should make truth traceable, action bounded, memory accountable, authority distributed, and harm continuously reviewable. That is what Ma'at contributes.

Introduction: From Capability to Constitution

The most important mistake in AI discourse is the assumption that intelligence is the primary engineering problem. It is not. Intelligence is only one variable. The deeper problem is governance under conditions of scale, opacity, autonomy, and speed. Once AI systems are embedded into search, coding, cybersecurity, education, administration, logistics, medicine, finance, and public decision-making, they cease to be mere applications. They become infrastructure. And infrastructure is never governed safely by cleverness alone. It requires constitutional order. Current international and technical frameworks already point in that direction. NIST says AI risk must be managed across the lifecycle of AI systems and that generative-AI risk management should explicitly address governance, content provenance, pre-deployment testing, and incident disclosure. UNESCO states that AI can never replace

ultimate human responsibility and accountability, and that monitoring and evaluation should be continuous and proportionate to risk. OECD similarly requires mechanisms for human agency and oversight, along with transparency and responsible disclosure.

The urgency is no longer theoretical. NIST has now released a concept note for a critical-infrastructure profile of the AI RMF, and Anthropic's 2026 Project Glasswing explicitly frames frontier models as a new factor in securing the world's most critical software. OWASP's LLM guidance now treats prompt injection, insecure output handling, training data poisoning, supply-chain vulnerabilities, excessive agency, and overreliance as core operational risks. Taken together, these developments reveal a transition: AI is not just a product category; it is becoming part of the civilizational substrate on which economies, institutions, and daily life depend.

My argument is that Ma'at is needed in all AI infrastructure because it offers what contemporary AI ethics often lacks: a unified moral-operational logic. Most AI ethics today is additive. It bolts fairness onto one subsystem, transparency onto another, security onto a third, and human oversight onto a fourth. Ma'at, by contrast, begins from the proposition that order, truth, balance, justice, and right relation are not optional supplements. They are the condition of legitimate power. An AI system that is not constituted by those principles can still be fast, persuasive, and commercially useful. It cannot, however, be fully trustworthy.

I. What Ma'at Is, and Why It Matters Here

Whether one approaches the subject through museum curation or African-centered ethical theory, the semantic core of Ma'at is consistent. The British Museum summarizes Ma'at as the deified personification of order, balance, truth, and justice, and links it to the duty of maintaining order. Karenga's doctoral work, *Maat, The Moral Ideal in Ancient Egypt*, presents Ma'at as a comprehensive ethical ideal expressed in classical Egyptian texts and explicitly explores its contribution to modern ethical discourse. Ma'at, then, is not reducible to piety or sentiment. It names a civilizational commitment to truthful order, just relation, and disciplined self-regulation.

Within African-centered intellectual history, scholars such as Cheikh Anta Diop and Théophile Obenga insisted that ancient Egypt must be restored to African intellectual history rather than severed from it. One need not settle every historiographical dispute in order to see the relevance of their intervention here. The important point is methodological: they rejected the colonial habit of treating African thought as pre-philosophical or morally ornamental. Their work reopened the possibility that African civilizational concepts can serve as serious sources of modern institutional design. In that sense, invoking Ma'at in AI is not antiquarianism. It is a refusal to let modern technology inherit only Euro-American categories of law, safety, and governance.

This matters because AI has a worldview problem. It is commonly trained on vast mixtures of code, language, documents, and behavior without a coherent moral center. It can imitate norms, but it does not generate a stable hierarchy of obligations from within itself. That is why today's most advanced systems require external controls, post-hoc filters, policy wrappers, and extensive red-teaming. The model is not self-governing. At best, it is instrumentally useful inside a governance regime built elsewhere. Ma'at is needed precisely because it supplies the

missing regime.

II. Why Existing AI Ethics Is Necessary but Insufficient

Contemporary AI governance is far from empty. NIST, UNESCO, OECD, and OWASP have all contributed essential pieces. NIST provides lifecycle-oriented risk management. UNESCO grounds AI ethics in human rights, accountability, transparency, and continuous evaluation. OECD emphasizes human-centred values, explainability, and oversight. OWASP names concrete failure modes in deployed LLM systems. These are indispensable achievements.

But these frameworks are still mostly federal in the bureaucratic sense, not constitutional in the civilizational sense. They identify risks and recommend controls, yet they often remain fragmented. Security is discussed here; fairness there; transparency somewhere else; provenance elsewhere; oversight somewhere else again. What is missing is a single governing principle that can tell an organization, at every layer of the stack, what kind of system it is morally authorized to build. Ma’at supplies that missing coherence.

To say this more sharply: current AI governance often asks, “What must we do to make this system safer?” Ma’at asks the prior question: “What kind of order is this system creating, and is that order truthful, balanced, just, and accountable?” The difference is profound. The first question is managerial. The second is constitutional. Managerial controls can reduce harm while leaving the underlying logic of domination, opacity, or irresponsibility intact. Constitutional order forces the system to justify its structure before it is ever deployed.

That is why Ma’at is not just “ethics for AI.” It is infrastructure doctrine. It asks whether the data pipeline tells the truth about its provenance; whether the model is deployed within proper limits; whether the orchestration layer respects proportionality; whether memory preserves accountability; whether tools operate under bounded authority; whether logs, audits, and incident disclosure create public intelligibility; and whether the human remains morally responsible rather than hiding behind machine outputs. Those are infrastructure questions, not branding questions.

III. The Case for Ma’at at Every Layer of AI Infrastructure

1. Data and training

AI systems are only as sound as the conditions under which their knowledge is assembled, labeled, filtered, and updated. NIST’s generative-AI profile explicitly highlights content provenance as a primary concern and notes that generative AI creates risks that differ from or intensify traditional software risks across the lifecycle, including ecosystem-level effects. OWASP lists training data poisoning and supply-chain vulnerabilities as core risks in LLM systems. If truth is not built into the provenance layer, the model inherits distortion at scale.

A Ma’at-governed data layer would therefore require provenance as a constitutional feature, not an optional metadata add-on. Every major dataset would need source lineage, method lineage, contamination tracking, confidence levels, contradiction flags, and bounded claims about representativeness. In Ma’at terms, this is the infrastructure of truth. Without it, AI is epistemically affluent but morally bankrupt: it knows many things, but cannot honestly say how it came to “know” them.

2. Model development and evaluation

NIST's current framework already treats AI risk management as something that must govern, map, measure, and manage risk across design, development, deployment, operation, and decommissioning. UNESCO further insists that ultimate responsibility and accountability must always remain with natural or legal persons, not with AI systems themselves. These are not minor compliance details. They imply that model evaluation cannot be reduced to benchmark performance. It must include questions of accountability, explainability, provenance, misuse potential, and downstream governance.

Ma'at sharpens this insight by refusing the cult of intelligence-for-its-own-sake. A model is not good because it is large, fast, or persuasive. It is good only insofar as its capabilities remain inside a truthful and balanced order. In other words, Ma'at subordinates model prowess to constitutional legitimacy. That is crucial in an era when frontier models are increasingly treated as cybersecurity-relevant infrastructure. Anthropic's Project Glasswing and NIST's critical-infrastructure work both signal that the question is no longer whether advanced AI can affect critical systems; the question is how to govern that fact before governance failures become systemic.

3. Orchestration and agentic behavior

OWASP warns against insecure output handling and excessive agency. The first means model outputs can trigger downstream exploits if not validated. The second means granting unchecked autonomy to LLMs can jeopardize reliability, privacy, and trust. This is precisely where Ma'at becomes indispensable. A Ma'at-governed system would not permit any model to act as sovereign. The model may propose, summarize, retrieve, classify, or recommend, but it does not become the final authority over execution.

Practically, this means that agentic systems require role separation, policy gates, and constrained execution. Discovery should not be the same as judgment; judgment should not be the same as persistence; persistence should not be the same as execution. The moral intuition here is ancient but technologically urgent: power should be distributed, not collapsed into a single opaque actor. Ma'at converts that intuition into architecture. Scout, analyst, archivist, guard, memory, and tool layers are not arbitrary components; they are institutional embodiments of a deeper principle that no unchecked intelligence should directly determine reality.

4. Memory, retrieval, and RAG

Modern AI infrastructure depends increasingly on external memory: vector stores, retrieval layers, summaries, agent logs, workspaces, and long-term state. Yet this is one of the least constitutionally governed layers of the stack. Systems often write vague summaries into memory, retrieve decontextualized fragments, and present them with unwarranted confidence. UNESCO emphasizes transparency, traceability, and the ability to challenge AI-informed decisions; NIST's generative-AI profile foregrounds content provenance; OECD likewise requires meaningful information about data sources and logic where feasible and useful. These requirements already imply that memory cannot be treated as a casual dumping ground.

A Ma'at-governed memory layer would therefore require that every durable record preserve who wrote it, from what source, by what method, under what confidence, and with what

contradiction status. Memory would become accountable rather than merely persistent. Retrieval would be obligated to show whether a result is canonical, provisional, contested, or anomalous. In Ma’at terms, memory would cease to be a convenience feature and become a truth-bearing institution.

5. Tools, plugins, and external action

The danger of modern AI is not only bad text. It is machine-mediated action. OWASP highlights insecure plugin design, excessive agency, and insecure output handling because contemporary models increasingly sit in front of tools, APIs, shells, code interpreters, databases, and operational systems. Once models can act, governance failures stop being merely epistemic and become material.

Ma’at is needed here because it transforms actuation into a governed privilege rather than a default capability. Every tool invocation should carry identity, scope, justification, policy review, and event logging. High-impact actions should require human approval or at least a stronger constitutional gate. This is not paranoia. It is simply the infrastructure translation of justice and balance: power should only be exercised within rightful bounds, in proportion to risk, and with traceable accountability.

6. Monitoring, redress, and institutional learning

UNESCO states that monitoring and evaluation of AI systems and AI ethics policies should be carried out continuously, systematically, and proportionately to relevant risks, with broad stakeholder participation and developed indicators. NIST’s profile also foregrounds incident disclosure as a primary consideration. These are signs that AI governance is shifting away from one-time certification and toward ongoing institutional self-examination.

That shift is, in essence, a Ma’at move. A Ma’at-governed system is not one that imagines itself perfect. It is one that preserves the capacity for self-correction under standards of truth and balance. In practical terms, that means anomaly tagging, event schemas, audit trails, post-incident review, redress channels, and constitutional revision when evidence shows the current order is failing. Ma’at here is not static morality. It is disciplined equilibrium.

IV. Why Ma’at Is Better Than a Generic “Ethics Layer”

A common objection is that all of the above could be achieved with generic AI ethics, without invoking Ma’at specifically. But this misses the point. The value of Ma’at is not that it contributes one more word to the ethics vocabulary. Its value is that it binds together truths often treated separately: epistemic truth, institutional order, moral balance, procedural justice, and accountable action.

Most ethics layers are advisory. Ma’at is constitutional. Most ethics layers sit above the stack. Ma’at belongs inside the stack. Most ethics layers ask teams to “consider” impacts. Ma’at asks them to structure power correctly before the system can act. That is a fundamentally different posture.

There is also a deeper reason. UNESCO’s Recommendation explicitly notes the importance of recognizing, protecting, and promoting endogenous cultures, values, and knowledge in AI governance, especially because many regions have been underrepresented in global AI ethics debates. A Ma’at-centered approach therefore does not represent a retreat from universality. It

represents a more honest universality: one that admits global AI cannot be normatively legitimate if it draws only from a narrow civilizational archive.

In this sense, Ma'at is not needed because Africa should be symbolically "included." It is needed because modern AI governance itself is impoverished when it excludes a mature African tradition of ordered truth, justice, proportion, and self-scrutiny. The question is not whether AI can survive without African ethical resources. It can, in the narrow sense that it can still operate. The question is whether it can become truly trustworthy, intelligible, and legitimate without them. My answer is no.

V. Objections and Replies

One objection says Ma'at is too culturally specific to ground "all" AI infrastructure. The reply is straightforward: every governance system is culturally specific at origin. Liberalism, utilitarianism, deontology, constitutionalism, and human-rights discourse all emerged from particular histories before being generalized. The real issue is not whether Ma'at comes from a specific civilization; it is whether its principles can govern plural systems without arbitrariness. Truth, balance, order, justice, accountability, and right relation are not parochial engineering needs. They are universal requirements for trustworthy socio-technical systems.

A second objection says modern AI already has NIST, OECD, UNESCO, and other frameworks, so Ma'at is redundant. The reply is that Ma'at is not redundant; it is integrative. Those frameworks already identify the right clusters of concern. Ma'at explains why they belong together and how they should be embodied institutionally. Put differently: modern frameworks provide many of the correct controls; Ma'at provides the civilizational logic that turns controls into a constitution.

A third objection says talk of moral foundations is impractical in engineering. Yet the opposite is now evident. AI security and governance bodies are increasingly focused on provenance, incident disclosure, oversight, supply chains, and bounded autonomy precisely because "just ship it" engineering has proven insufficient. Anthropic's Project Glasswing, NIST's critical-infrastructure profile work, and OWASP's risk taxonomy all testify that governance is now an engineering problem, not an after-dinner philosophical concern.

Conclusion

The central claim of this dissertation is now clear. Ma'at is needed in all AI infrastructure because advanced AI is not merely computation; it is organized power. And organized power, if it is to remain legitimate, must be truthful, ordered, balanced, just, accountable, and self-correcting. Those are precisely the terms Ma'at names.

NIST, UNESCO, OECD, and contemporary AI security practice have already converged on many of the practical necessities: governance across the lifecycle, provenance, human responsibility, transparency, explainability, oversight, continuous evaluation, incident disclosure, and secure deployment. What they have not yet fully named is the deeper unity of those requirements. Ma'at names that unity.

Therefore the proper future of AI is not merely "aligned models." It is Ma'at-governed infrastructure: data with provenance, models under bounded authority, memory with accountability, tools behind policy gates, agents under role separation, monitoring tied to redress, and human beings who remain the final bearers of responsibility. A system built this

way does more than comply. It becomes constitutionally legible. It can be questioned, corrected, trusted, and limited.

That is why Ma'at is not decorative language for AI. It is the missing constitutional grammar of the AI age.

End of manuscript draft. A full chaptered dissertation outline with citations, thesis statement, literature review, methodology, and bibliography may follow.